

An Efficient Miniaturized Tri-band THz Antenna for Future 6G Systems and Beyond

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Abstract—This manuscript delineates the design and performance of a wideband microstrip patch antenna, envisioned for utilization in the imminent 6G era and beyond, specifically tailored for terahertz communication systems. The antenna is meticulously crafted with a star-shaped structure, positioned on a Rogers RT5880 substrate, accompanied by a partial slotted ground plane. The antenna exhibits operation in three distinct bands accomplishing -10 dB impedance bandwidths of 500 GHz, 730 GHz, and 1280 GHz, respectively. Furthermore, the antenna demonstrates peak gains (6.95 dB) and high radiation efficiency (95%). The antenna boasts a commendable voltage standing wave ratio (VSWR) of approximately 1.1 indicative of excellent impedance matching with the load. The proposed antenna possesses the merits of extensive bandwidth, high efficiency, and miniaturized size, rendering it apt for terahertz wireless communication networks such as wearables and Internet of Things (IoT) sensors, making it well-suited for a wide range of applications in the 6G and beyond ecosystem.

Keywords—Tri-band; THz antenna; 6G communication; IoT, Rogers RT5880.

I. INTRODUCTION

The rapid advancement of wireless communication technologies has prompted the exploration of terahertz (THz) frequencies as a key enabler for future 6G systems. Emerging applications such as holographic telepresence, extended reality, and ultra-high-speed wireless backhaul are fueling an unprecedented demand for data traffic. To address this surge, wireless systems must tap into new frequency bands that offer ultra-wide bandwidths, with the THz band (0.1 to 10 THz) being a promising candidate [1]. Its vast unused frequency range and high spatial resolution make it essential for next-generation mobile networks, though it also presents challenges in antenna design and performance. As 5G networks are deployed globally, it's essential to begin envisioning 6G, which is expected to harness the THz band for unprecedented wireless services [2]. While 5G mm-Wave communications require 13.5 GHz of spectrum, this may not meet the growing bandwidth demand. Higher frequencies, like THz, offer vast spectral resources suitable for Tbps communications. The IEEE 802.15.3d standard supports data rates of 100 Gbps for short-distance THz transmission [3]. By dispersing the signal

across a broad frequency range, spread spectrum techniques make it considerably more difficult for potential eavesdroppers to intercept the signal. Nonetheless, it's important to recognize that THz communications present unique security challenges that necessitate specific solutions. IoT is anticipated to grow exponentially in the coming decade, with 500 billion connected devices by 2030 and a market value exceeding USD 6 billion by 2025 [4]. To accommodate this rapid expansion, a new generation of wireless technology will be essential, offering ubiquitous coverage, high performance, and diverse functionalities to meet the increasing demand. 6G is envisioned as such a technology, as it will offer enhanced communication capabilities, such as ultra-reliable low-latency communications (URLLC), enhanced mobile broadband (eMBB), and massive machine-type communications (MMTC), as well as wireless sensing, which will play a critical role in improving the service quality and user experience of IoT systems [5]. Global initiatives are advancing 6G networks for IoT applications, such as 6Genesis in Finland, Hexa-X in Europe [6], and the FCC in the U.S. which testing 6G features on mobile and IoT devices [7]. South Korea plans a 6G pilot project in digital healthcare and smart cities starting in 2026, with commercialization projected by 2030 [8]. Researchers are also exploring advancements in wireless networks, particularly in THz frequencies.

This review summarizes recent progress, trends, challenges, and potential solutions for future 6G systems, focusing on the integration of these technologies for enhanced IoT capabilities. One of these efforts is the study in [9] by A. Mir et al., which proposes a compact cross bow tie antenna for the terahertz range, offering circular polarization without a complex feeding network. It shows a return loss > -10 dB, a 50% 3 dB axial ratio bandwidth, and symmetric radiation patterns from 100-600 GHz. In the works presented by authors A. Sharma et al. [10] presents the simulation results of a rectangular microstrip patch antenna for 0.7–0.85 THz, achieving a 17.76% impedance bandwidth and 3.5 dB gain, with good omnidirectional radiation and 55% efficiency, ideal for short-distance wireless communication. The paper [11] presents a terahertz antenna with 16 sets of air holes in a thick silicon substrate, improving return loss (-67.32 dB), gain (9.75 dB), and achieving a 47 GHz bandwidth at 0.607 THz. J. O. Abolade presents a miniaturized semicircular multiband

antenna for terahertz WBAN applications, optimized at 0.18 THz with a size of $400 \times 400 \times 5 \mu\text{m}^3$, offering peak gains of 4.5 dB, 4.1 dB, and 5.5 dB across three bands [12]. A CP monopole antenna for THz communication, fabricated on 127 μm Rogers RT6006, resonates at ~ 0.20 and ~ 0.56 THz with a VSWR of 1.4, covering 0.19–0.24 and 0.27–0.61 THz bands [13]. For multiple THz applications this paper [14] introduces a half-moon patch antenna on a PBG substrate, operating at 0.63 THz with 10.5 dBi gain, 60 GHz bandwidth, -23 dB return loss, and a VSWR of 1.1 by I. Ahmad et al. The authors S.K. Vijay et al. [15] analyze five THz microstrip patch antennas with a modified PBG substrate, showing minimal return loss of -83.73 dB and 90.84% radiation efficiency. Their large $700 \times 900 \mu\text{m}^2$ EBG fractal triband antenna operates at 0.948, 0.984, and 1.040 THz for THz applications. The work in [16] demonstrates that fractal dimensions improve bandwidth (8, 40, 20 GHz) while maintaining good impedance matching, with a reflection coefficient under -10 dB and a VSWR below 2. In [17], denser PBG structures ($500 \times 500 \times 200 \mu\text{m}^3$) are shown to enhance the gain (7.99 dB) of a THz antenna, with a graphene-based patch antenna on an Arlon (AR-100) substrate achieving high radiation efficiency (91.1%) at 1.04 THz for wireless communication applications.

Previous work on 6G IoT systems has faced limitations in achieving efficient performance across multiple frequency bands, especially in the THz spectrum. Many existing antenna designs struggle with narrow bandwidth, poor radiation efficiency, and suboptimal performance at higher frequencies. In contrast, this paper introduces a novel tri-band terahertz antenna with a partial ground plane that achieves -10 dB bandwidths at 500, 730, and 1280 THz at resonant frequencies of 1.77, 3.2, and 5.2 THz. This star-shaped circular slot antenna, made from copper, offers impressive return loss, optimal voltage standing wave ratio, substantial gain, and up to 85% radiation efficiency, addressing key challenges in THz applications. A detailed performance evaluated in section III and finally a comparison, future work and conclusion are presented in section IV, V and VI respectively.

II. ANTENNA DESIGN

This section presents the geometry and parameters of the proposed THz antenna. The antenna consists of a star-shaped patch with ten irregular arms and a circular parasitic element at the center, placed on a 13.38 μm thickness of Rogers RT5880 substrate ($\epsilon=2.2$, $\tan\delta=0.0009$) with a dimension of 300×300 . The patch and ground element having a thickness of 2.0 μm and are made of copper conductivity ($\sigma = 5.8 \times 10^7 \text{ S/m}$). The patch is fed by a microstrip edge line feeding technique with a width of 22 μm and result is analyzed by CST Microwave Studio simulation software. The design evaluation of the THz antenna, as shown in Fig. 1, is broken down into distinct evaluation stages. Initially, the design begins with a basic rectangular shape (top-left), which represents the first stage in the antenna's layout. As the design progresses, the antenna's geometry is refined into a star-shaped configuration with a central hole (top-right), showcasing a more complex and functional structure that is suited for THz frequencies. The bottom stages highlight the final assembly where the antenna design is tested and integrated with additional features, such as the base, shown in the bottom-left quadrant. The final bottom-left section depicts the final phase of the process, which includes both physical creation and functional evaluation. A u-shaped slotted partial ground plane is used to achieve the desired tri-band frequency. This methodical progression from design to fabrication ensures the antenna meets the required

specifications for THz applications. Fig. 2(a, b) shows a front and back view of the proposed antenna respectively. The optimized dimensions in Fig. 2 and the values of each parameter that define the antenna are summarized in Table I.

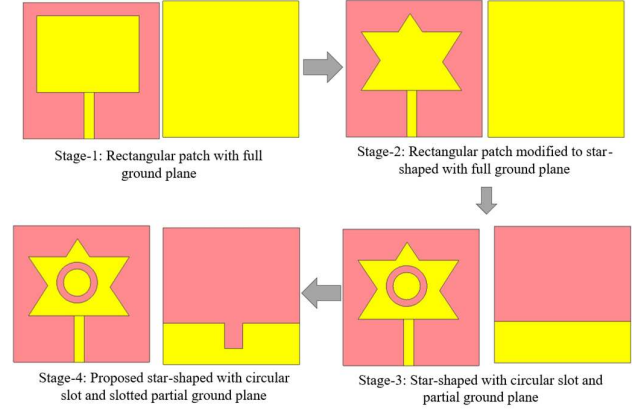


Fig. 1. Progressive design evaluation of proposed THz antenna.

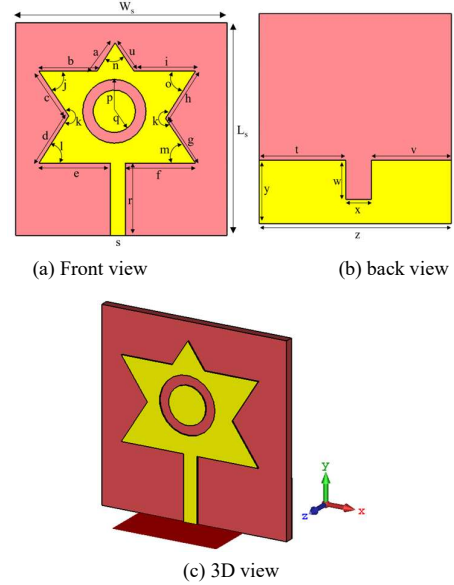


Fig. 2. Proposed star-shaped THz antenna.

TABLE I. PHYSICAL DIMENSIONS OF PROPOSED ANTENNA

Symbol	Dimension (μm , °)	Symbol	Dimension (μm , °)
Substrate width (W_s)	300	Lower left angle (l)	57.51
Substrate length (L_s)	300	Lower right angle (m)	56.09
Apex left arm (a)	47.42	Apex angle (n)	66.40
Upper horizontal left arm (b)	82.53	Upper right angle (o)	57.51
Upper angular left arm (c)	75.97	Slot radius (p)	45
Lower angular left arm (d)	78.90	Parasitic radius (q)	30
Left base arm (e)	101	Feeder length (r)	101.90
Right base arm (f)	99	Feeder width (s)	22
Lower angular right arm (g)	75.97	Left edge ground plane (t)	135
Upper angular right arm (h)	78.90	Right edge ground plane (v)	125
Upper horizontal right arm (i)	87.11	Ground plane slot height (w)	56
Apex right arm (u)	48.20	Ground plane slot width (x)	40
Upper left angle (j)	56.09	Ground plane height (y)	91
Upper-lower segment angle (k)	246.40	Ground plane width (z)	300

III. RESULTS AND PARAMETRIC INVESTIGATION

A. S-parameter

The proposed antenna is a wideband antenna that can operate in the terahertz (THz) frequency range shown in Fig. 2, which is suitable for high-speed wireless communication

and beyond. The reflection coefficient for all five stages of the THz antenna design is illustrated in Fig. 3. Proposed design of stage-4 offering optimized performance compared to the earlier stages. The antenna consists of a star-shaped radiating element and feed with line edge technique that maintain good impedance. The circular-shaped slot and u-slotted partial ground plane are used to create multiple resonant modes and enhance impedance bandwidth operated in range under 1.5-2 THz, 2.98-3.71 THz and 4.74-6.02 THz band. The reflection coefficient (S11) is used to assess the antenna's wideband performance, indicating how well it matches the transmission line. A value below -10 dB ensures minimal power reflection, and simulations using CST software calculate this coefficient for accurate performance evaluation. The simulation results of the reflection coefficient vs. frequency are shown in Fig. 4. We conclude that the proposed antenna has three resonance at 1.77 THz upper-band, 3.21 THz mid-band and 5.2 THz lower-band, where the reflection coefficient reaches the minimum values of -26.29 dB, -23.64 and -24.71 dB respectively. These values are much lower than -10 dB, which means that the antenna is well matched to the transmission line at these frequencies. The antenna cover three frequency region including lower-bandwidth about 1280 GHz, mid-band region 730 GHz and upper band of 500 GHz (S11 < -10 dB), which is suitable for various THz applications.

The Voltage Standing Wave Ratio (VSWR) is a measure used in the field of antenna design to evaluate the impedance matching condition. It is defined as the ratio of the maximum to the minimum voltage along the transmission line. Fig. 5 gives an indication of the amount of power that is reflected back from the antenna due to a mismatch between the source and antenna impedances in terms of VSWR. A lower VSWR value implies a better impedance match, meaning that most of the power is delivered to the antenna. A common criterion for a satisfactory impedance matching value of $1 \leq \text{VSWR} \leq 2$. This corresponds to a reflection coefficient of <-20 dB. In the case of the proposed antenna, the VSWR plot (shown in Fig. 5) demonstrates that the VSWR value is around 1.14 (≤ 2) within three operating band of 1.5-2 THz, 2.98-3.71 THz and 4.74-6.02 THz. The minimum VSWR values of 1.10, 1.14 and 1.1 are achieved at the resonant frequencies of 1.77, 3.21 and 5.23 THz respectively. This confirms the excellent impedance matching performance of the proposed antenna. Therefore, the proposed antenna can show excellent wideband performance in THz regime.

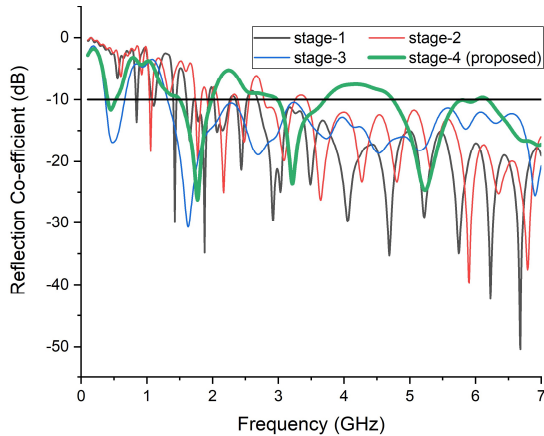


Fig. 3. Reflection co-efficient for different stages of the proposed antenna.

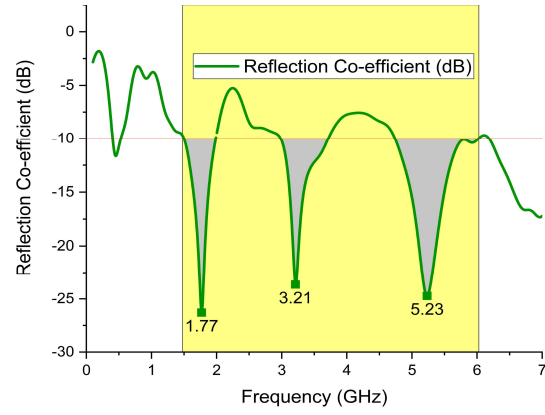


Fig. 4. S-parameter of proposed star-shaped THz antenna.

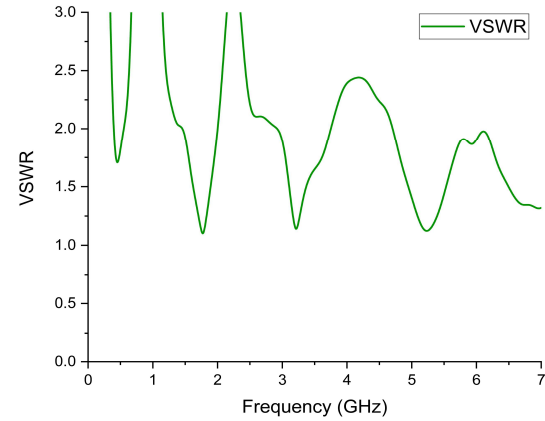


Fig. 5. VSWR of proposed star-shaped THz antenna.

B. Gain and Directivity

The gain and directivity of the proposed Terahertz antenna are investigated and plotted in Fig. 6 as a function of frequency. According to the figure, the gain of Fig. 6 at resonance frequency is 4.28 dB, 3.54 dB and 1.24 dB in the lower, middle and upper bands from 1.5-2 THz, 2.98-3.71 THz and 4.74-6.02 THz respectively. The directivity characteristics over the operating band obtain a peak value of 6.93 dB at 3.55 THz in mid-band as shown in Fig. 6.

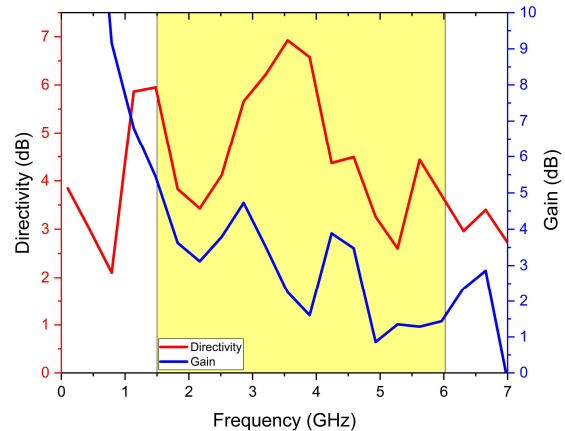


Fig. 6. Gain and directivity of proposed star-shaped THz antenna.

C. Radiation pattern and radiation efficiency

The radiation patterns of a terahertz antenna are depicted in Fig. 7. These patterns are presented in polar form and correspond to different frequencies: 1.77 THz, 3.2 THz and 5.2

THz. Each frequency corresponds to a unique combination of electric field (E) and magnetic field (H) components, as well as their azimuth angle (ϕ). At a frequency of 1.77 THz Fig. 7(a), the main lobe of the electric and magnetic fields is directed at 21° when $\phi=0^\circ$ and 124° when $\phi=90^\circ$, respectively. Interestingly, there are no side lobes in either field at this frequency when $\phi=90^\circ$. The maximum half-power beamwidth for both E and H fields at a frequency of 4.6 THz shows 66.8° ($\phi=0^\circ$), 46.4° ($\phi=90^\circ$) in Fig. 7(b). The main lobe peak magnitude at $\phi=90^\circ$ is 21.2 dBV/m at an angle of 159° when the magnetic field value is -30.3 dBA/m. Notably, there is a side lobe present in both field directions at this frequency. In 5.2 THz resonance frequency Fig. 7(c) E, H field radiated at angle of 81° , 134° for $\phi=0^\circ$, 90° respectively. Fig. 8 depicts the three-dimensional radiation patterns of the antenna at its corresponding resonant frequencies. As illustrated in Fig. 9,

the proposed tri-band antenna exhibits a high degree of radiation efficiency for terahertz applications, making it a viable choice for various applications.

The radiation efficiency of the antenna, which ranges from 85% to 90% distributed across three bands: 1.5-2 THz, 2.98-3.71 THz, and 4.74-6.02 THz. At resonance frequency in the upper band at 90%, followed by the mid-band at 87%, and the lower band at 85%. The radiation characteristics of a terahertz antenna vary significantly with frequency shows maximum about 95% at 1.82 THz, demonstrating the complex interplay between the electric and magnetic fields and their azimuth angles. This complexity underscores the importance of understanding these patterns for the effective design and operation of terahertz antennas.

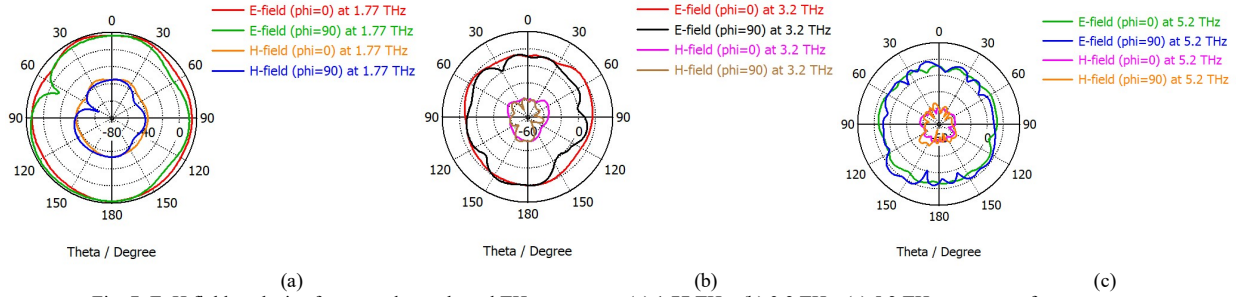


Fig. 7: E, H field analysis of proposed star-shaped THz antenna at (a) 1.77 THz, (b) 3.2 THz, (c) 5.2 THz resonance frequency.

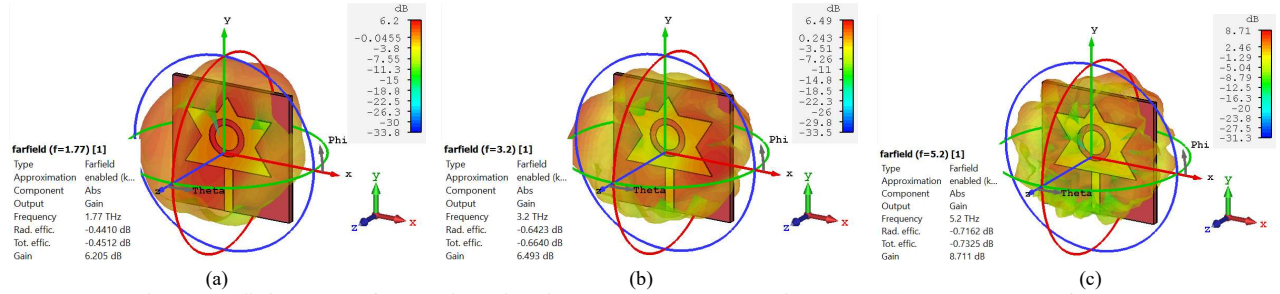


Fig. 8: 3D radiation pattern of proposed star-shaped THz antenna (a) 1.77 THz, (b) 3.2 THz, (c) 5.2 THz resonance frequency.

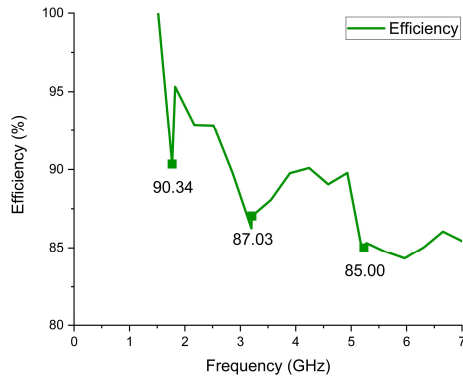
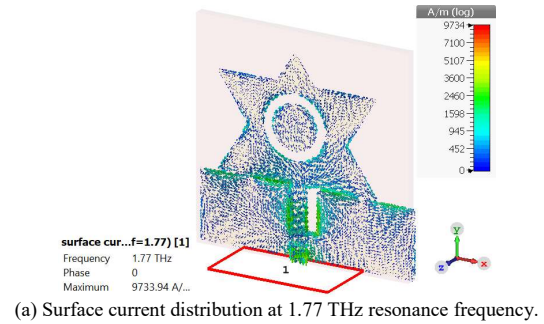


Fig. 9. Radiation efficiency of proposed star-shaped THz antenna.

D. Surface current distribution

The distribution pattern of the current is depicted in Fig. 10, showcasing three distinct resonance frequencies across three separate frequency bands (1.5-2 THz, 2.98-3.71 THz, 4.74-6.02 THz). As observed in Fig. 10, the surface current is primarily concentrated at the lower patch and circular edge of the patch structure and significantly influences the partial ground plane at a frequency of 5.2 THz.



(a) Surface current distribution at 1.77 THz resonance frequency.

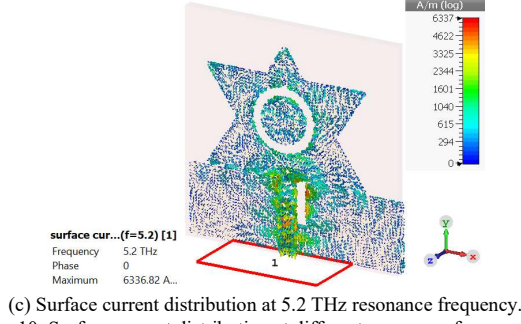
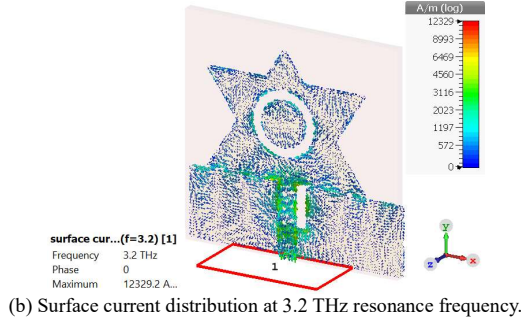


Fig. 10: Surface current distribution at different resonance frequencies.

IV. COMPARATIVE ANALYSIS

The performance of the proposed antenna has been benchmarked against other comparable terahertz antennas, as detailed in Table II. However, it was found that several research studies lacked crucial parameters. For instance, the material used for the patch was not specified in studies [9], [10], [11], [12], [14], and [15]. Furthermore, efficiency and gain were omitted in studies [18], [12], [24], and [16]. Despite the presence of certain gaps, Table II clearly demonstrates that the proposed antenna exhibits optimal performance across several parameters including its dimensions, large bandwidth, efficiency, and gain. This suggests that it holds significant potential for integration into contemporary miniaturized systems for future generations.

V. FUTURE WORK

Future research will include the integration of the antenna into full communication systems, testing its performance in real-time data transfer, and evaluating its compatibility with 6G technologies such as massive MIMO, beamforming, and THz signal processing. Fabrication and measurement of the antenna will be conducted to validate the simulation results, ensuring real-world performance aligns with theoretical predictions. Exploration of alternative substrates with higher dielectric constants or lower loss factors, such as graphene and other advanced materials, will be investigated to further improve performance.

TABLE II: COMPARISON WITH RECENT RELEVANT WORK

Ref.	Size (L×W×h) μm ³	Substrate material	Operating frequency range (THz)	BW (GHz)	Reflection coefficient (dB)	Centre frequency (THz)	Max Gain (dB)	Max Efficiency (%)	Application
[9]	1600×1600×100	Quartz	0.1-0.6	500	~-21	0.11, 0.312, 0.54	10.43	-	THz-band Represented
[10]	1000×1000×200	RT/Duriod6 006	0.7-0.85	150	~-41	0.775	3.5	55.88	Short Distance Wireless Communication Systems
[11]	500×500×50	Silicon-air photonic crystal	0.583-0.63	47	-67.32	0.607	9.75	89.79	Proposed a THz design
[12]	400×400×5	Rogers RO3210	0.17-0.2, 0.49-0.56, 0.68-0.99	30, 70, 310	~-28.0; ~-17.0; ~-19.0	0.18; 0.53; 0.71, 0.96	4.5, 4.1, 5.5	99.98	Wireless body area network, imaging, IoT
[13]	850×770×197	Rogers RT6006	0.19-0.24, 0.27-0.61	50, 340	~-15.1, ~-48	~0.20, ~0.56	5.8	-	THz Communication
[14]	700×660×192	polyimide	~0.61-0.65	40	30.66	~0.63	9.45	82.6-90.6	THz Applications
[15]	600×600×90	Polyimide	0.5-0.8	300	-83.73	0.61	9.19	90.84	Sensing and Communication Technologies
[16]	900×700	RT Duriod (6010)	~0.942-0.95, ~0.98-0.99, ~1.02-1.04	8, 40, 20	~-16.5, ~-26, ~-15	0.948, 0.984, 1.04	9	-	THz application
[17]	500×500×200	Arlon (AR-100)	~1.2-1.3	100	-24	1.04	7.99	91.1	Size-limited wireless applications
Proposed work	300x300x13.38	Rogers RT5880	1.50-2.00, 2.98-3.67, 4.73-5.58,	500, 690, 850	-26.29, -23.64, -24.71	1.77, 3.21, 5.23	6.95	95	Future 6G Systems and Beyond

VI. CONCLUSION

A wideband high efficiency microstrip patch antenna has been presented and investigated in range between 1.5-2 THz, 2.98-3.71 THz, 4.74-6.02 THz spectrum. The proposed star-shaped structure of THz antenna has been modified upon

300 × 300 μm² Rogers RT5880 substrate with a partial ground plane improved -10 dB impedance bandwidth of 500 GHz, 730 GHz and 1280 GHz. The simulation results reveal that antenna has gain of 4.28, 3.54, 1.24 dB respect to its resonance frequency on each separate band. The obtained high gain

makes the antenna promising for communication systems. The model also achieved radiation efficiency up to 85% and simulated in CST software shows VSWR of around 1.1 indicating good impedance matching with load and exhibits good reflection co-efficient < -20 dB. The improved broad bandwidth, high radiation efficiency and miniaturized multiple

resonance characteristics of our proposed antenna are particularly advantageous for terahertz communication systems in the forthcoming 6G era and beyond.

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